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Research Project Report on the Topic:

Analysis of Factors Influencing Financial Indicators

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Аннотация

Данная работа посвящена эконометрическому анализу структурных изломов в поведении индекса Московской биржи (MOEX) в период с января 2020 по по середину 2024 года. Это был один из наиболее тяжёлых периодов в истории российского финансового рынка. Основными источниками внешних шоков выступили пандемия COVID-19, введение масштабных экономических санкций, а также геополитическая эскалация 2022 года.

Для анализа использовались три взаимосвязанных блока: краткосрочная OLS-модель с коррекцией HAC, процедура Бай-Перрона для обнаружения структурных изломов и долгосрочны анализ коинтеграции с помощью метода динамического МНК (DOLS). Данные охватывают 1102 торговых дня по индексу MOEX и включают такие факторы, как цены на нефть Brent, индекс доллара США (DXY), доходность 10 летних казначейских облигаций США и индекс волатильности VIX.

По результатам анализа все три выдвинутые гипотезы были подтверждены: наличие структурного излома в краткосрочных зависимостях (подтверждено тестом Чоу), существование условной долгосрочной коинтеграции в постшоковом периоде, а также фундаментальная нестабильность долгосрочных коэффициентов (зафиксирована тестом CUSUM). После 2022 года чувствительность MOEX к нефтяным ценам ослабла почти на 70%, в то время как роль обменного курса усилилась более чем в два раза.

Полученные результаты свидетельствуют о смене режима функционирования российского фондового рынка.

Abstract

This study is devoted to the econometric analysis of structural breaks in the behavior of the Moscow Exchange Index (MOEX) over the period from January 2020 to mid-2024. This was one of the most challenging periods in the history of the Russian financial market. The primary sources of external shocks were the COVID-19 pandemic, the imposition of large-scale economic sanctions, and the geopolitical escalation of 2022.

The analytical framework consists of three interconnected blocks: short-term OLS regression with HAC correction, the Bai-Perron procedure for detecting structural breaks, and long-term cointegration analysis using the Dynamic Ordinary Least Squares (DOLS) method. The dataset covers 1102 trading days of the MOEX index and includes key factors such as Brent crude oil prices, the US Dollar Index (DXY), the yield on 10 year US Treasury bonds, and the CBOE Volatility Index (VIX).

The results of the analysis confirm all three research hypotheses: the presence of a structural break in short-term relationships (confirmed by the Chow test), the existence of conditional long-term cointegration in the post-shock period, and the fundamental instability of long-term coefficients (detected by the CUSUM test). After 2022, the sensitivity of the MOEX index to oil prices decreased by almost 70%, while the role of the exchange rate increased more than twofold. The obtained results indicate a regime shift in the functioning of the Russian stock market.

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Key Terms

Structural breaks: statistically significant changes in the parameters of an econometric model or time series process, indicating shifts in the underlying economic relationships over time.

MOEX Index: the main capitalization-weighted stock market index of the Moscow Exchange reflecting the performance of the largest and most liquid Russian companies.

Bai-Perron procedure: an econometric methodology for detecting and estimating multiple structural breakpoints in linear regression models using global optimization and sequential testing algorithms.

Cointegration: a statistical property of non-stationary time series that move together in the long run, forming a stable equilibrium relationship despite short-term deviations.

Dynamic OLS (DOLS): an estimator of cointegrating relationships that corrects for endogeneity and serial correlation by including leads and lags of first differences of regressors.

Russian stock market: the system of organized trading of equity securities in Russia, primarily represented by the Moscow Exchange and its benchmark indices.

Brent crude oil: a global benchmark grade of crude oil used for pricing international oil markets and serving as a key external determinant for commodity-exporting economies.

USD/RUB exchange rate: the nominal exchange rate expressing the value of the Russian ruble relative to the US dollar, reflecting external balance and capital flow dynamics.

VIX: the CBOE Volatility Index measuring market expectations of future volatility in the US stock market and commonly interpreted as a global risk sentiment indicator.

Regime shifts: transitions between distinct periods characterized by different statistical properties, behavioral patterns, or economic mechanisms in financial time series.

Machine Learning (ML): a set of data-driven computational methods that allow models to identify patterns, detect nonlinearities, and improve predictive performance without explicit parametric specification.

Data collection: the systematic process of gathering, validating, and organizing raw financial and macroeconomic time-series data for subsequent empirical analysis.

Introduction

The period from 2020 to 2025 represents one of the most unstable periods in economic history, affecting the entire world. The COVID-19 pandemic, disruptions in global supply chains, the imposition of widespread economic sanctions, and especially geopolitical tensions have significantly changed the dynamics of financial markets, especially in emerging markets. These events have created conditions in which well-established, stable economic relations have undergone significant structural changes.

During this period, the Russian stock market stood out significantly, experiencing pronounced volatility reflecting both external shocks and internal economic changes. Given the structural complexity of trade in Russia, which for a long time was almost entirely dependent mainly on commodity exports, and has been actively changing over the past almost 20 years (since the 2008 crisis), it is very important to assess whether the traditional factors determining stock market dynamics have remained stable or have changed in magnitude and direction. Understanding such changes is important for assessing the stability of the market, investors, long-term equilibrium relations, and the broader process of economic adaptation to extreme conditions.

Although many of the studies I have found deal in detail with structural shifts in financial time series, and others with various crisis events, their isolation from each other and a limited number of studies do not allow for a comprehensive econometric assessment of successive shocks within a single analytical system. In particular, there is insufficient data on whether the long-term relationship between the MOEX index and its key macrofinancial determinants has fundamentally changed.

The purpose of this study is to identify and analyze potential structural changes in the behavior of the Moscow Stock Exchange index in the period 2020-2025. Whether the relationship between the Moscow Stock Exchange index and its main determinants has changed during this period and how the sensitivity to these determinants has changed in different periods is what interests me in this study.

To answer this question, the study will use the Bay-Perron multiple structural discontinuity procedure in combination with cointegration analysis using the dynamic least squares method (DOLS). This approach will make it possible to identify shifts in the regime and assess long-term equilibrium ratios, taking into account correlation and other factors.

This study aims to provide a systematic econometric assessment of the structural transformations in the Russian stock market in the context of a number of major external shocks, rather than focusing on a single isolated event or market area. By evaluating regime-dependent parameters, the study will provide a quantitative assessment of how economic sensitivity has changed over time.

To analyze the existing scientific literature, the next section (literature review) examines both the methodological foundations for identifying structural disorders and empirical studies examining the impact of recent shocks on the Russian economy. The rest of the article is organized as follows: section 2 provides an overview of the literature, section 3 describes the dataset and methodological framework, the work plan is listed in section 4, and the final section is devoted to the bibliography, the materials on which I rely during my research.

Literature Review

Bai, J., & Perron, P. (2003). Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1), 1-22.

Description. This article is a classic and fundamental work in econometrics, focusing on the identification and analysis of multiple structural breaks in linear regression models. The authors extend the theoretical results presented in their earlier work (Bai & Perron, 1998) and propose computational algorithms, confidence interval construction methods, and testing and selection procedures for the number of breaks.

Connection to the topic. The Bai and Perron method is the basic, generally accepted standard for analyzing complex periods like 2020-2025 in the Russian market, where one can expect not one but several breaks (Covid-19 pandemic, 2022 and subsequent adaptation phases). This work provides a rigorous statistical toolkit that will be central to the practical aspects of my research. It will serve as a foundation for further investigation.

Methods and models. The authors develop an algorithm based on dynamic programming for efficient global search of breakpoints. They present tests for the presence of breaks (supF test), double maximum tests (UDmax, WDmax) to determine the number of breaks, and use information criteria (Bayesian Information Criterion, LWZ) to select the optimal number of segments.

Results. The Bai-Perron procedure proves highly efficient and consistent in both simulation studies and practical research. It successfully identifies the precise moments in time when the relationships in the data - such as the sensitivity of stock returns to oil prices - fundamentally change. This makes it a reliable tool for detecting structural breaks in financial time series.

Zeileis, A., Leisch, F., Hornik, K., & Kleiber, C. (2002). strucchange: An R Package for Testing for Structural Change in Linear Regression Models. *Journal of Statistical Software*, 7(2), 1-38.

Description. This article introduces the strucchange package for R, which is used to test whether the relationships in a statistical model change over time. It includes popular tests such as CUSUM and MOSUM, which track the fluctuations of model errors, as well as F-tests (such as the Chow test). The package allows to run these tests, create graphs to visualize the potential change points with boundaries, and obtain a statistical assessment of their significance. The key feature is that it can also track new data as it becomes available, using methods such as moving estimates to alert user to changes as soon as they occur.

Connection to the topic. For the practical implementation of the analysis stated in the topic, a convenient and reliable software tool is necessary. The `strucchange` package allows the application of Chow, Quandt-Andrews (QLR) tests and visualization of results (e.g., using CUSUM plots), which is critically important for the empirical stage of the work. Its integrated functions for model diagnostics and structural break detection provide a complete analytical workflow, streamlining the transition from statistical theory to actionable results.

Methods and models. The methods and models implemented in the package fall into two main frameworks. The first is the F-test framework, which includes the Chow test for a known breakpoint and the Quandt-Andrews (supF/QLR) test for an unknown single breakpoint. The second is the generalized fluctuation framework, featuring tests based on recursive residuals (CUSUM) and recursive/moving coefficient estimates, complemented by dedicated functions for graphical visualization.

Results. The package provides researchers with a full set of tools for screening and detecting structural changes, greatly simplifying the technical side of the analysis. The ability to use it in academic and applied research enhances its reliability and versatility.

Montalvo, J. G. (1995). Comparing cointegrating regression estimators: Some Monte Carlo evidence. *Economics Letters*, 48(3-4), 229-234.

Description. This article conducts a Monte Carlo study to compare the finite sample properties of various estimators used in cointegrated regression models. The main focus is on evaluating the canonical cointegration regression (CCR) method and the dynamic ordinary least squares (DOLS) method proposed by Stock and Watson in 1993, compared to the standard ordinary least squares (OLS) method. This study uses data collection mechanisms developed by Inder. B. in 1993 to assess efficiency under conditions of endogeneity and sequential correlation.

Connection to the topic. This work is of great importance for creating a reliable econometric base for the primary analysis within my research project. Before applying structural change tests to the relationship between the Moscow Exchange index and explanatory variables such as oil prices and the exchange rate, it is crucial for me to correctly identify and estimate any possible long-term equilibrium relationship. Using an inappropriate method can lead to false regression results and unreliable parameter estimates, and it can invalidate any subsequent analysis of structural breaks based on this model.

Methods and models. The methodology is based on Monte Carlo tests. The author generates artificial time series based on specific cointegrated systems with known parameters and then applies different estimators (OLS, CCR, DOLS) to recover those parameters. Performance is measured by statistical bias and Root Mean Squared Error (RMSE) across many test runs.

Results. The key finding is that the DOLS estimator systematically outperforms both the OLS and the CCR estimators in terms of having smaller bias and RMSE across most of the simulated data-generating processes, especially when short-run dynamics are present. This provides strong simulation-based evidence in favor of using the DOLS approach for estimating cointegrating relationships in applied work with limited sample sizes. In the future, it will be much easier for me to organize my analysis by dividing it into shorter segments.

Zaremba, A., Kizys, R., Aharon, D. Y., & Demir, E. (2020). Infected markets: Novel coronavirus, government interventions, and stock return volatility around the globe. *Finance Research Letters*, 35, 101597.

Description. This global study empirically examines the impact of the COVID-19 pandemic and subsequent government interventions on stock return volatility across 67 countries in early 2020. The authors analyze the correlation between the growth in confirmed coronavirus cases and various government responses (lockdowns and economic stimulus) with market stability and volatility of shares.

Connection to the topic. This work is crucial for analyzing the first major shock in 2020 of the Russian market. It provides a robust, globally comparative framework. This article shows that the Russian market likely experienced a significant structural break in its volatility regime in March 2020, aligning with the global trend of "infected markets," and the scale of this break may have been amplified by the simultaneous oil price shock, which is very relevant for Russia.

Methods and models. The primary model uses a random-effects estimator, with standard errors robust to heteroskedasticity and autocorrelation. Volatility is measured through five proxies: the logarithm of absolute daily returns and residuals from four asset pricing models (CAPM, Fama-French, Asness and Carhart). The key independent variable is the Government Response Stringency Index, alongside daily changes in COVID-19 infections and deaths.

Results. The study finds a strong, positive relationship between the growth in Covid-19 cases and increased stock market volatility globally. Government interventions showed a complex effect: while harsh containment policies initially increased volatility, economic support packages helped to mitigate it. This provides a direct benchmark for assessing the nature of the potential structural break in the Russian MOEX index's behavior during the initial pandemic phase.

Klose, J. (2023). Empirical effects of sanctions and support measures on stock prices and exchange rates in the Russia-Ukraine war. *Journal of International Financial Markets, Institutions and Money*, 59.

Description. This article, which focuses on the financial consequences of the first 13 months of the Russia-Ukraine conflict (February 1, 2022, to February 24, 2023), analyzes the impact of various types of sanctions and support measures on global markets. The key findings highlight the asymmetric effect of sanctions and the surprising neutrality of financial assistance for the financial market, allowing policymakers on both sides to develop economic measures during the war.

Connection to the topic. An ideal source for my research of structural breaks as it shows a significant event that occurred in February-March 2022. Using a panel VAR model, the author shows that financial sanctions (such as disconnection from SWIFT) have the most significant negative impact on the Russian stock market and lead to the devaluation of the ruble. This allows us to make a clear hypothesis about the nature and strength of the change in the behavior of the MOEX index.

Methods and models. Author used panel vector autoregression (Panel-VAR) with a breakdown of effects by type of sanctions (financial, trade and others) and by sending countries (especially the G7).

Results. It seems that for the first 13 months sanctions have had a pretty visible impact on Russia as a target country. The effect is most pronounced for financial sanctions and for sanctions imposed by G7 countries. Sanctions have also had a negative impact on Ukraine, which highlights the complexity of the cross-effects.

Yalcin, E., et al. (2025). On the effectiveness of the sanctions on Russia: New data and new evidence. *VoxEU/CEPR*.

Description. This article presents new research that is based on the Global Sanctions Database, revealing that the 2022 sanctions on Russia have had a complex and counterintuitive effect. While reducing Russia's trade with sanctioning countries, they simultaneously triggered a significant liberalization of trade with non-sanctioning nations like China and Turkey.

Connection with the topic. This source shows not just a "weakening of ties", but a fundamental restructuring of Russia's economic ties after 2022, which is the macroeconomic basis for a structural shift in the stock market. The study confirms that Russia has reoriented trade flows from the West to East. This explains why the influence of traditional global factors (G7 markets) on MOEX could weaken, while the role of internal and "friendly" factors could increase. This will have a very interesting effect on the economy that will be crucial to understand.

Methods and models. Author used a gravity-based trade model based on the Global Sanctions Database (GSDB-R4) and IMF/UN trade data.

Results. The primary effect of sanctions (reduced trade with senders) turned out to be moderate (about -25% on average). However, Russia compensated for this by significantly increasing trade with countries that did not join the sanctions. For example, trade with India has more than doubled since 2022.

Pan, Q., & Sun, Y. (2023). Changes in volatility leverage and spillover effects of crude oil futures markets affected by the 2022 Russia-Ukraine conflict. *Energy Economics*, 103.

Description. Analyzing high-frequency data, this study examines the impact of the 2022 geopolitical shock on oil market volatility. These data provide important information for risk management and portfolio strategy in energy markets during periods of intense geopolitical tension.

Connection with the topic. This paper directly answers the question about how exactly the oil-market channel could have been interrupted. The work proves that the conflict has fundamentally changed the behavior of the oil market: the leverage effect has changed, as well as the effects of spillover effects between major oil brands. This is direct evidence of a structural break in the key driver of the Russian market.

Methods and models. Author uses asymmetric GARCH models (GJR-GARCH, ADCC) for analyzing changes in volatility and correlation parameters before and after February 2022.

Results. The Russia-Ukraine conflict has significantly changed the leverage effect on the WTI, Brent and Oman oil futures markets. Dynamic conditional correlations have stabilized, and volatility between markets has weakened.

Atlantic Council (2025). The Russian economy in 2025: Between stagnation and militarization.

Description. This is a comprehensive report by a leading U.S. think tank that provides a detailed analysis of the structural state of the Russian economy three and a half years after the beginning of Ukraine conflict. The article summarizes and tries to predict the long-term effects of sanctions, wartime fiscal policy, and the fundamental reorientation of foreign trade toward China into a full picture of a current and future economy.

Connection to the topic. This report is basically a summary of 2020-2025 analysis. It provides the macroeconomic context for understanding the "new equilibrium" of the Russian stock market after the sequence of structural breaks (COVID-19, the 2022 sanctions shock, further adaptation). The report's conclusions on economic overheating, supply-side constraints (labor, investment), and the new dependence on China explain the fundamental environment in which the MOEX index now operates. This will help me to look at how economy adapts to such powerful structural breaks.

Methods and models. This analysis is based on macroeconomic data synthesis and trade flow analysis. It employs economic concepts such as "military Keynesianism" and examines imbalances in the Russia-China trade relationship, where Russia primarily exports raw materials and imports manufactured goods.

Results. The report finds that Western sanctions, while not causing economic collapse, have locked Russia into a militarized growth path dependent on China. It identifies key structural vulnerabilities, including an overheating economy, a hollowed-out civilian sector, and a reliance on war spending.

Data and Methodology

6.1 Data Description

The dataset I created and used in this study examines the structural changes in the behavior of the Moscow Stock Exchange Index (MOEX) for the period from January 1, 2020 to June 2024, covering 1101 trading days available through Yahoo Finance for the IMOEX.ME ticker. The dataset has a panel structure consisting of 26,617 daily observations across 18 international stock market indices, where each observation unit is identified by the combination of market name and trading date. Despite such structure, the primary analytical object is the MOEX index, for which 1,102 trading-day observations are available over the sample period.

The unit of observation is a single equity market on a single trading day with daily data frequency. Geographic coverage spans 18 markets across North America, Europe, Asia-Pacific, and emerging markets, providing a comparative framework for contextualising MOEX behavior relative to global peers.

The dependent variable is constructed in two forms corresponding to short-run and long-run analysis. Talking about the short-run specification, the daily logarithmic return of the MOEX index captures how the market reacted to major shocks on a day-to-day basis. As for the long-run specification, the closing level of the MOEX index is used to investigate potential cointegrating equilibrium relationships with macro-financial determinants.

All stock index data are sourced from Yahoo Finance via the Python library. Macro-financial variables - Brent crude oil futures (ticker BZ=F), the US Dollar Index (DX-Y.NYB), the CBOE Volatility Index (VIX), and the 10-year US Treasury yield (TNX) - are also sourced from Yahoo Finance at daily frequency. The COVID-19 Government Stringency Index for Russia is obtained from the Oxford COVID-19 Government Response Tracker (OxCGRT).

The structure of the final MOEX-specific analytical dataset is confirmed by `moexdata.info()`, the output of which is summarised in table below. The dataset contains 1,101 observations across 15 columns. The `Date` column is stored as `datetime64[ns]`, the `market` identifier as `object`, and all 13 remaining variables as `float64`. A null-count check confirms zero missing values across every column, verifying data integrity prior to modelling. The first-differenced variables (`d_Brent`, `d_DXY`, `d_VIX`, `d_US10Y`, `d_COVID`) are included here as they appear after preprocessing - they are used exclusively in the short-run OLS specification. Memory usage amounts to 137.6 KB, while the date range spans from 2020.01.06 to 2024.06.14.

Table 1: MOEX analytical dataset structure

#	**Column**	**Non-Null**	**Dtype**
0	Date	1101	`datetime64[ns]`
1	market	1101	`object`
2	Close	1101	`float64`
3	returns	1101	`float64`
4	volatility_21d	1101	`float64`
5	VIX	1101	`float64`
6	Brent	1101	`float64`
7	DXY	1101	`float64`
8	US10Y	1101	`float64`
9	COVID_Stringency	1101	`float64`
10	d_Brent	1101	`float64`
11	d_DXY	1101	`float64`
12	d_VIX	1101	`float64`
13	d_US10Y	1101	`float64`
14	d_COVID	1101	`float64`
Memory usage: 137.6+ KB. Zero null values across all columns.			

6.2 Variables Definition

The table below lists all variables used in the study with their descriptions, integration order, measurement units, and roles across model specifications.

Table 2: Variable definitions, integration orders, and model roles

Variable	**Description**	**Order**	**Unit**	**Role**
Close (MOEX)	Closing value of the MOEX index	I(1)	Index points	Dependent - long-run
returns (r_t)	Daily log return: $\ln(MOEX_t/MOEX_{t-1})$	I(0)	Dimensionless	Dependent - short-run
volatility_21d	21-day rolling std of log returns	I(0)	Dimensionless	Descriptive only
Brent	Brent crude oil futures price	I(1)	USD/barrel	Explanatory - commodity
DXY	US Dollar Index (trade-weighted)	I(1)	Index	Explanatory - exchange rate
VIX	CBOE Volatility Index	I(0)	Index points	Short-run model only
US10Y	10-year US Treasury yield	I(1)	Percent	Explanatory - monetary
COVID_Stringency	Russia COVID-19 Stringency Index	I(0)	Score 0-100	Short-run model only
ΔBrent	First difference of Brent	I(0)	USD/barrel	Short-run OLS regressor
ΔDXY	First difference of DXY	I(0)	Index	Short-run OLS regressor
ΔUS10Y	First difference of US10Y	I(0)	Pct. points	Short-run OLS regressor

ADF unit root tests (more about them is described in Section 4.4) determine each variable's integration order. The only macro-financial variable that has been verified as I(0) (ADF $p = 0.001$) is VIX, which has significant ramifications for model specification because it is only utilized as an exogenous control in the short-run OLS specification and cannot be included in the cointegrating vector in the long-run DOLS model. Because COVID-Stringency is bounded and has policy-driven dynamics rather than random-walk dynamics, it is also considered as I(0). In accordance with Stock & Watson (1993), the I(1) variables (MOEX level, Brent, DXY, and US10Y) enter the DOLS cointegrating regression in levels, and their first differences with one lead and one lag function as endogeneity correction terms.

6.3 Data Preprocessing

Thus, the following preprocessing steps were performed, each of which was necessary to work correctly with the dataset.

Forward-filling of macro-financial variables. The macro-financial series sourced from Yahoo Finance and OxCGRT operate on different trading calendars than MOEX. Missing values arising from calendar misalignments - for example, US public holidays coinciding with Russian trading days - were filled using the last available observation (`.ffill()`). This approach is standard practice for daily financial panel data and ensures that the most recent available information is used without introducing any look-ahead bias.

Rolling volatility initialisation. The 21-day rolling standard deviation of returns requires a minimum of 21 prior observations. To avoid missing values at the start of the 2020 analysis window, data collection begins from January 2019, providing a sufficient burn-in period. Only observations from January 1, 2020 onward are used in all modelling stages.

Exclusion of trading volume. The trading volume variable for the MOEX index (ticker IMOEX.ME) contained exclusively zero values - a known limitation of Yahoo Finance for index-level tickers, which do not report exchange-aggregated volume. The variable was excluded from all model specifications and is acknowledged as a data limitation in Section 6.7.

Extreme observations retained. No outlier removal, winsorisation, or smoothing was applied to return observations. The -40.5% single-day return on 24 February 2022 and the elevated volatility cluster of March 2020 are genuine market events central to the research question. Removing them would distort the structural breaks that the study aims to detect.

First-difference construction. First differences of I(1) regressors (ΔBrent, ΔDXY, ΔUS10Y) are computed as $\Delta X_t = X_t - X_{t-1}$. For the DOLS specification, one lag and one lead of each first difference are additionally constructed to serve as endogeneity and serial-correlation correction terms. The construction of leads and lags reduces the effective DOLS sample to $\mathrm{N} = 1{,}098$.

Final dataset check. After all preprocessing, the MOEX-specific analysis dataset contains 1,101 observations with zero missing values across all variables, verified by explicit null-check (`isna().sum().sum() = 0`).

6.4 Descriptive Statistics

The table below presents summary statistics for all key variables in the MOEX analysis dataset.

Table 3: Descriptive statistics for key variables (MOEX dataset, N = 1,101)

Variable	**Mean**	**Std**	**Min**	**25%**	**Median**	**75%**	**Max**
MOEX (Close)	3012.2	579.5	1917.0	2526.0	3079.5	3439.1	4287.5
returns	0.000040	0.01990	−0.405	−0.00514	0.00125	0.00701	0.183
volatility_21d	0.01360	0.01460	0.00305	0.00756	0.00970	0.01397	0.106
VIX	21.72	8.41	11.86	16.20	20.09	25.51	82.69
Brent	74.26	20.94	19.33	63.21	78.52	86.24	123.58
DXY	99.52	5.91	89.44	93.55	100.39	104.16	114.11
US10Y	2.525	1.385	0.515	1.302	2.642	3.850	4.988
COVID_Stringency	31.36	20.70	0.00	11.11	31.02	47.69	87.04

A number of characteristics of these data deserve consideration. The yield distribution on MOEX is very unusual: the one-day collapse on February 24, 2022 is represented by a low of −0.405, and the erratic behaviour typical of emerging markets in the face of a geopolitical shock is reflected in the excess kurtosis of 171.1, which is confirmed by the Jarque–Bera test in Section 4.6. Clustering of strong volatility is confirmed by the 21-day moving average volatility, which reaches a maximum of 0.106 during the crisis in February–March 2022, while its average value is 0.010. The price of Brent crude oil ranges from \$19.33 at the time of the negative territory incident in April 2020 to \$123.58 at the time of the supply cutoff in March 2022. The 10-year US Treasury yield represents a complete monetary policy cycle within the sample, ranging from 0.515% (COVID-era rates close to zero, 2020) to 4.988% (Federal Reserve tightening cycle, 2023).

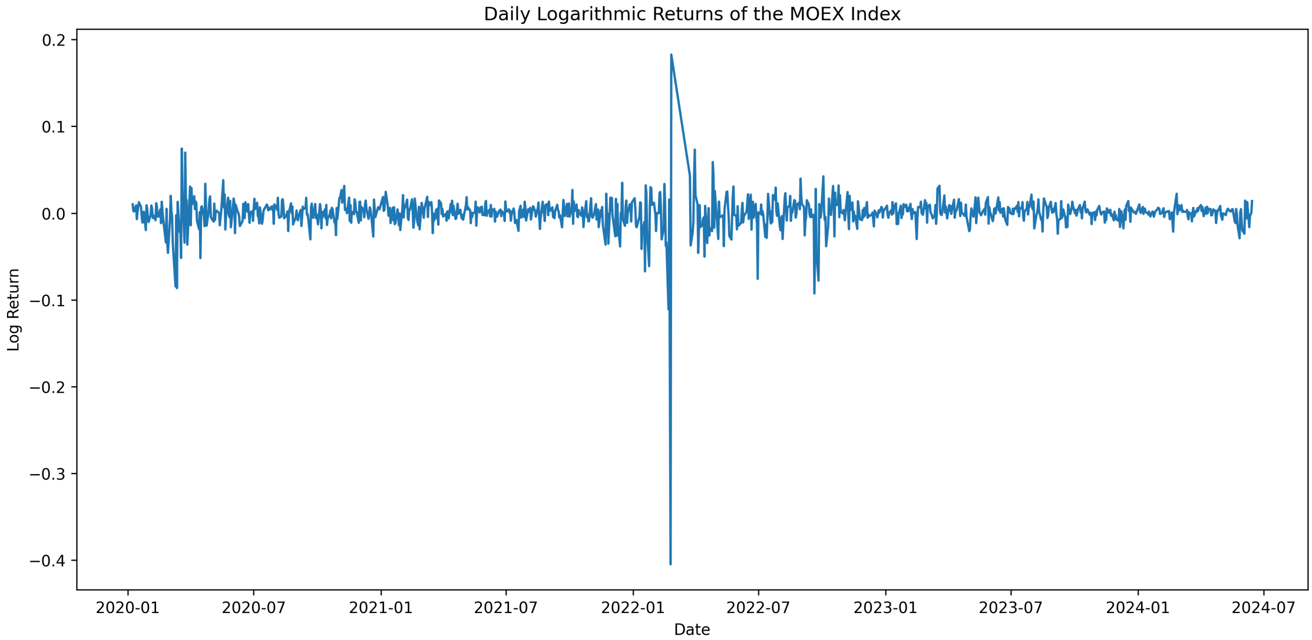

Figure 1: Daily logarithmic returns of the MOEX index (2020-2024).The extreme −40.5% observation on 24 February 2022 is clearly visible. Volatility clustering around March 2020 and February-March 2022 are motivators for my analysis of structural breaks.

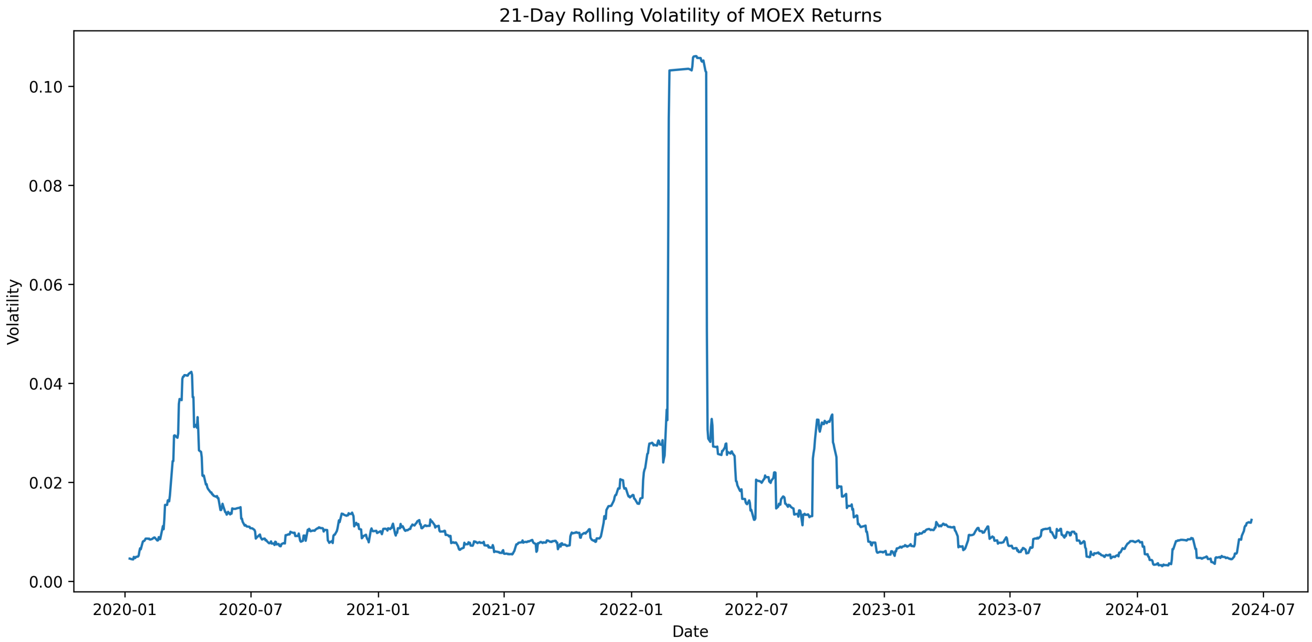

Figure 2: 21-day rolling volatility of MOEX returns. Two distinct volatility spikes correspond to the COVID-19 shock (March 2020) and the geopolitical shock (February 2022). Volatility returns to low levels by late 2022, indicating a new stable regime.

Significant multicollinearity patterns between the explanatory variables are shown by the correlation matrix. DXY–US10Y (r = 0.82), Brent–US10Y (r = 0.69), and Brent–DXY (r = 0.55) have the strongest inter-regressor relationships. On the other hand, all explanatory variables exhibit nearly zero correlations with MOEX daily returns ($|r| < 0.10$), indicating a limited short-run linear dependence between macro-factors and MOEX returns. The use of Ridge regularization in the ML baseline (to address multicollinearity) and HAC standard errors in OLS (to address residual autocorrelation and heteroskedasticity) are two methodological decisions that I made motivated by this pattern, which is in line with the findings of Zaremba et al. (2020) for emerging markets during crisis periods.

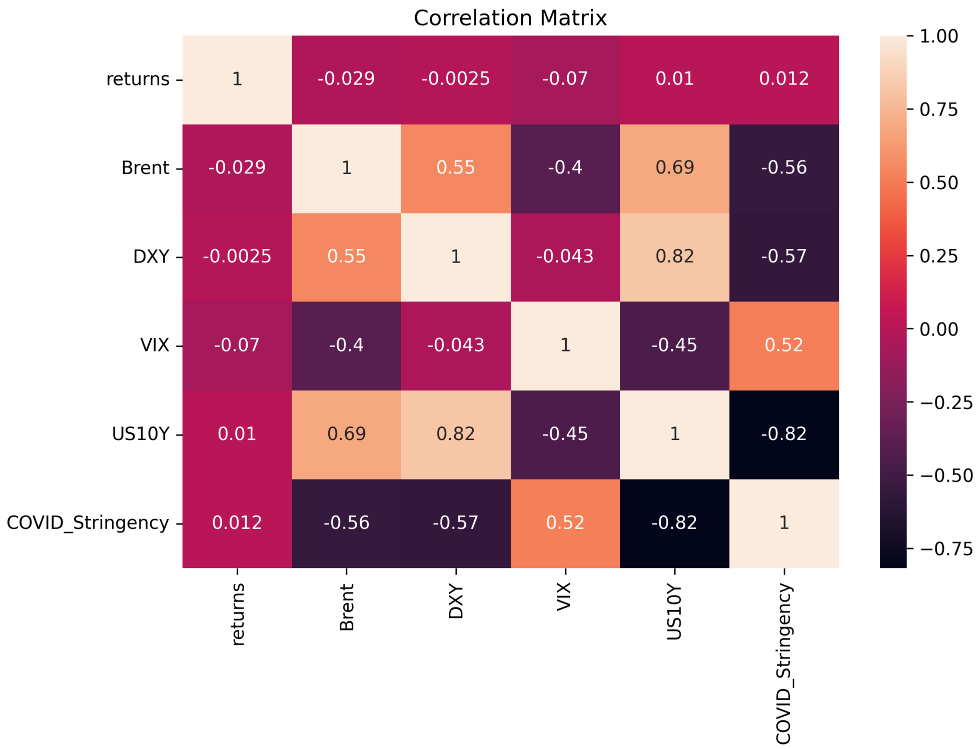

Figure 3: This is a correlation matrix of the main variables. Near-zero correlations between returns and all regressors confirm limited short-run linear dependence. Strong DXY-US10Y ($r = 0.82$) and Brent-US10Y ($r = 0.69$) inter-regressor correlations motivate Ridge regularisation in ML specifications.

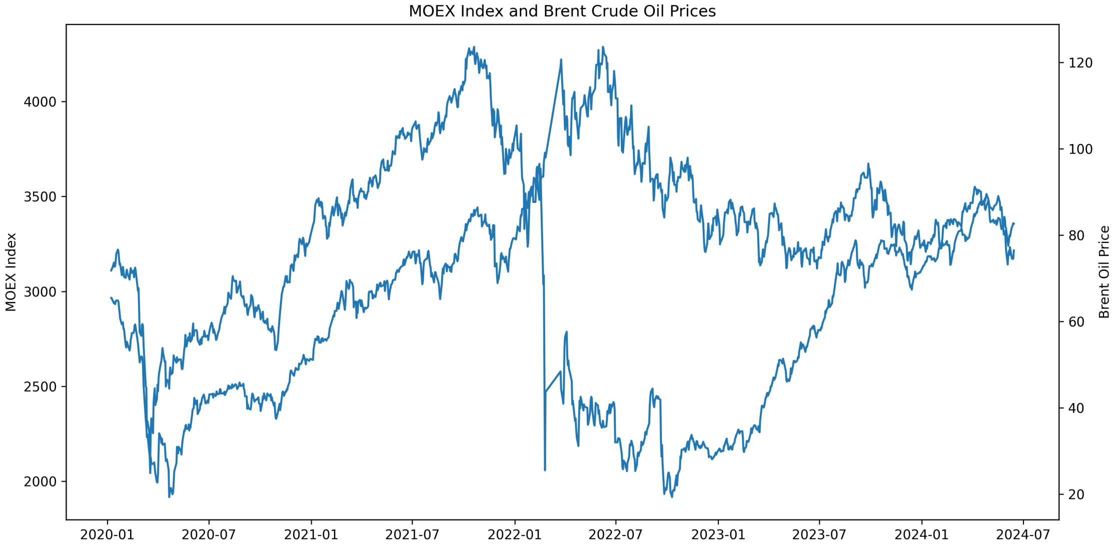

Figure 4: This is a MOEX index level and Brent crude oil prices (dual axis, 2020-2024). Both series trend together over the long run despite short-run divergence around the 2022 shock, visually motivating the cointegration analysis of Hypothesis 2 (more detailed in 4.5).

6.5 Research Hypotheses

Hypothesis 1 (Short-term Sensitivity)

H_0: The coefficients of the regression of MOEX returns on oil (Brent), VIX, DXY and other factors are stable over the entire period 2020-2025 (no structural breaks exist).

H_1: At least one of the coefficients has significantly changed at one or more points in time (there exists at least one structural break).

Significance level: $\alpha = 0.05$.

Test: Bai–Perron multiple structural break procedure applied to OLS residuals via the PELT algorithm. Chow test at the known break date of 24 February 2022.

Justification for the choice of test: The Bai-Perron (2003) strategy is selected as the main method since a data-driven approach is necessary and the number and timing of breaks are unknown in advance due to the research period's several subsequent shocks (COVID-19, sanctions, mobilisation). The process, which synthesises previous contributions such as CUSUM tests (Brown, Durbin & Evans, 1975) and QLR/supF tests (Andrews, 1993), is the methodological gold standard for multiple break detection. Following Klose's (2023) identification of February 24, 2022, as the major structural event for Russian financial markets, the Chow test is employed as a confirmatory technique for this particular economically motivated date.

Economic interpretation: If H_1 is supported, it means that the sensitivity of MOEX returns to oil price changes, dollar movements, and global risk indicators shifted structurally after beginning od the conflict.Under sanctions and capital controls, the transmission mechanisms linking global macro-factors to Russian equity returns were fundamentally altered - consistent with the market isolation documented in Klose (2023) and the trade reorientation described in Yalcin et al. (2025).

Hypothesis 2 (Long-term Equilibrium)

$\mathbf{H_0}$**:** The level series of MOEX, Brent, DXY, VIX and US10Y are not cointegrated - there is no long-term equilibrium, and the index follows a pure random walk relative to the macro-factors.

$\mathbf{H_1}$**:** There exists a stationary long-term linear combination between MOEX and the set of macro-financial variables (cointegration).

Test: Engle-Granger two-step procedure - Dynamic OLS (DOLS) estimation of the cointegrating vector, followed by an ADF test on the residuals to verify stationarity.

Justification for the choice of test: DOLS was chosen in response to Montalvo (1995), who used Monte Carlo simulation to show that, in finite samples, DOLS consistently performs better than both simple OLS and the Canonical Cointegrating Regression (CCR), achieving lower bias and RMSE, especially when short-run dynamics are present. The estimator, which was first put out by Stock & Watson (1993), adjusts for endogeneity by adding leads and lags of the regressors' first differences to the cointegrating regression. This is significant in the context of MOEX because MOEX fluctuations may simultaneously affect oil prices and the rouble exchange rate through feedback pathways. Since VIX is verified to be I(0) and cannot be included in a legitimate cointegrating connection with I(1) variables, it is not included in the cointegrating vector.

Economic interpretation: If H_1 is supported, it confirms that the Russian stock market maintains a long-run fundamental relationship with world commodity prices, the dollar, and global monetary conditions - meaning that MOEX deviations from equilibrium are temporary and mean-reverting. If H_0 cannot be rejected, the MOEX level is driven by random processes and is not anchored to fundamental macro-financial variables, which would have significant implications for long-run investors and policymakers.

Hypothesis 3 (Structural Instability of Long-term Relationships)

$\mathbf{H_0}$**:** The long-term DOLS coefficients (γ_1 Brent, γ_2 DXY, etc.) are stable over the entire period 2020-2025 - the cointegrating relationship does not change.

$\mathbf{H_1}$**:** The long-term coefficients undergo significant structural breaks under the impact of COVID and/or geopolitical shocks of 2022-2025.

Significance level: $\alpha = 0.05$.

Test: Bai-Perron procedure applied to DOLS residuals (PELT algorithm, penalty parameter $p = 30$, confirmed stable at $p = 50$), with additional confirmation through CUSUM recursive residual tests.

Justification for the choice of test: The Bai-Perron procedure is applied to DOLS residuals because it is specifically designed for detecting multiple breaks in linear and cointegrating regressions (Bai & Perron, 2003). Supplementary CUSUM tests implement the classic recursive residual framework of Brown, Durbin & Evans (1975), operationalised via the `recursive_olsresiduals` function in `statsmodels`. This combination provides both formal multiple-break detection and visual recursive diagnostics, following the analytical workflow described in Zeileis et al. (2002).

Economic interpretation: If H_1 is confirmed, a post-2022 weakening of the Brent coefficient would reflect the decoupling of MOEX from global oil benchmarks due to the Urals discount. A strengthening of the DXY coefficient would reflect the dominant role of the exchange rate channel under capital controls. A sign reversal in the US10Y coefficient would indicate a fundamental change in the transmission of global monetary conditions to the Russian equity market under financial isolation.

6.6 Statistical Methods

In this empirical analysis, six complementary methodological approaches are applied, following the structure: Data → Preprocessing → Descriptives → Hypotheses → Models → ML Baseline → Limitations.

1. Short-term OLS regression and diagnostic tests

The short-run dynamics of MOEX daily returns are analysed using the following specification:

$$r_t = \alpha + \beta_1 \Delta Brent_t + \beta_2 \Delta DXY_t + \beta_3 VIX_t + \beta_4 \Delta US10Y_t + \beta_5 COVID_t + \varepsilon_t$$

I(1) regressors (ΔBrent, ΔDXY, ΔUS10Y) enter as first differences to ensure stationarity. VIX and COVID_Stringency enter in levels as I(0) variables. OLS estimation is applied with HAC (Newey-West) standard errors, `maxlags` = 5. The justification for HAC is grounded in three diagnostic tests:

The **Breusch-Pagan test** (LM = 15.19, $p = 0.010$) confirms heteroskedasticity, expected given volatility clustering in the MOEX return series. The **Durbin-Watson statistic** (DW = 2.262) indicates no first-order autocorrelation. The **Jarque-Bera test** (JB = 1,304,830, $p \approx 0$) confirms severe non-normality (kurtosis = 171.1), driven by the -40.5% observation of 24 February 2022. These results collectively confirm that HAC standard errors are both necessary and appropriate across all OLS specifications.

The short-run OLS yields $R^2 = 0.042$, consistent with near-efficient-market behavior in emerging markets. Only ΔBrent ($p = 0.032$) and ΔDXY ($p = 0.029$) are statistically significant at $\alpha = 0.05$.

2. Structural break analysis - Bai-Perron, Chow, CUSUM

The Bai-Perron procedure (PELT algorithm, `ruptures` library) is applied to OLS residuals with penalty $p = 5$ and $p = 10$. Two breakpoints stable across both penalty values are identified: 2021-11-18 and 2022-10-28, corresponding to the onset of geopolitical escalation and the post-mobilisation adaptation period respectively. The Chow test at 24 February 2022 yields F = 4.853, $p = 0.000067$ ($k = 6$, $N_1 = 536$, $N_2 = 564$), confirming a statistically significant structural break at the 95% confidence level. **H_1 is supported.**

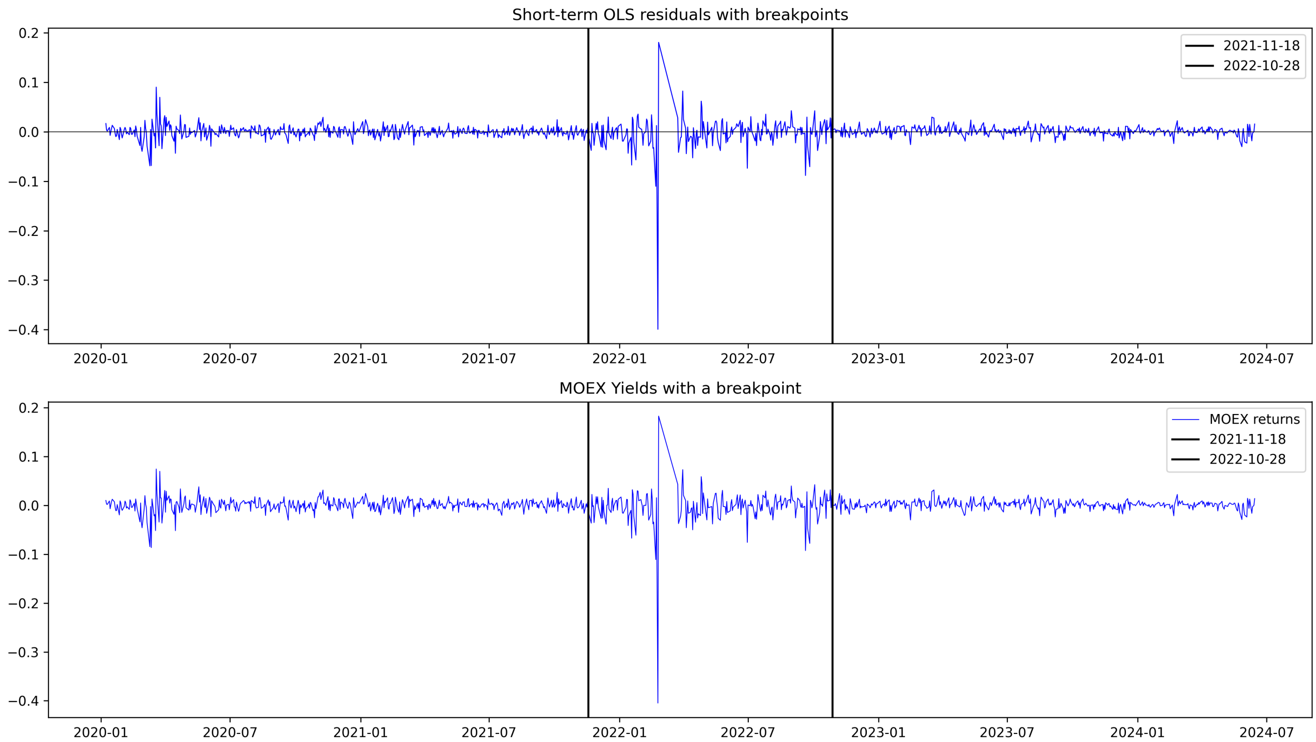

Figure 5: Short-term OLS residuals with structural breakpoints (top) and MOEX daily returns with breakpoints (bottom). Vertical lines mark the data-driven breakpoints at 2021-11-18 and 2022-10-28. The visible change in residual structure around these dates confirms parameter instability in the short-run model.

3. DOLS cointegration analysis

Since MOEX, Brent, DXY, and US10Y are confirmed I(1), the Engle-Granger two-step cointegration procedure is applied. The long-run equilibrium relationship is estimated via DOLS following Montalvo (1995):

$$MOEX_t = \gamma_0 + \gamma_1 Brent_t + \gamma_2 DXY_t + \gamma_3 US10Y_t + \sum_{j=-1}^{1} \delta_j \Delta X_{t-j} + u_t$$

VIX is excluded as I(0). One lead and one lag ($k = 1$) are included as endogeneity correction terms. HAC standard errors with `maxlags` = 5 (full sample) and `maxlags` = 3 (sub-periods) are applied.

On the full sample, the ADF test on DOLS residuals yields statistic = −2.643 ($p = 0.085$) and Durbin-Watson = 0.025 - the near-zero DW signals extreme positive autocorrelation, indicating that the full-sample model mixes two structurally different data-generating processes. Sub-period estimation at the Bai-Perron break date of 2022-02-16 reveals: (a) pre-break period shows spurious regression ($p = 0.244$, $R^2 = 0.870$ with non-stationary residuals) (b) post-break period confirms cointegration ($p = 0.000$). **H_2 is conditionally confirmed for the post-break period.**

All three long-run coefficients changed significantly ($p < 0.05$) across sub-periods:

Table 4: Sub-period DOLS long-run coefficient comparison

Variable	**Pre-break**	**Post-break**	**Change**	**Direction**
Brent	+28.042	+8.335	−70.3%	Weakened
DXY	−28.195	−85.299	−202.5%	Strengthened
US10Y	−193.937	+733.198	Sign reversal	Fundamental change

4. Bai-Perron on DOLS residuals and CUSUM

The Bai-Perron procedure applied to DOLS residuals (pen = 30, stable at pen = 50) identifies four breakpoints: 2020-07-13, 2021-06-18, 2022-02-16, and 2023-08-10. The primary break at 2022-02-16 - eight days before the invasion - is consistent with the forward-looking nature of financial markets.

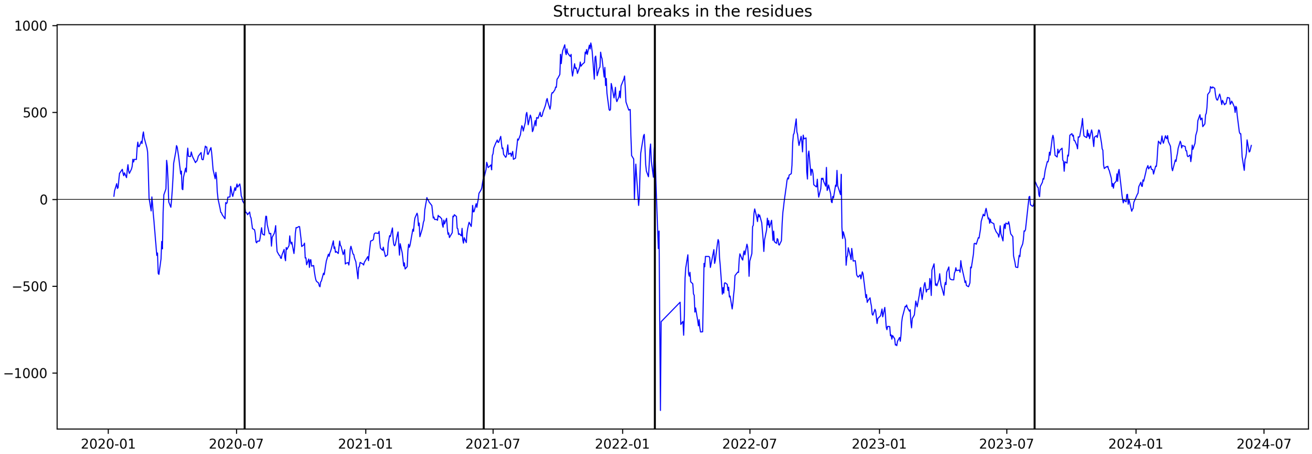

Figure 6: Structural breaks in DOLS residuals (2020-2024). Four vertical lines mark the Bai-Perron breakpoints. The systematic shift in residual level after 2022-02-16 confirms regime change in the long-run equilibrium relationship.

The CUSUM recursive residual test confirms parameter instability: the statistic exceeds the 5% critical boundaries, formally rejecting H_0 of coefficient stability. **H_3 is confirmed.**

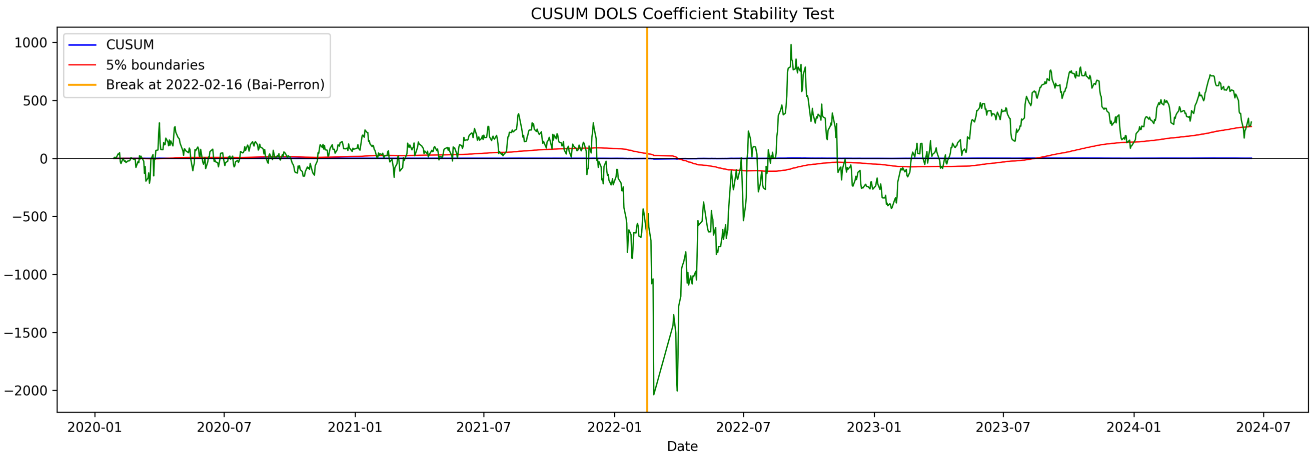

Figure 7: CUSUM test for DOLS coefficient stability (H_3). The blue line shows the CUSUM statistic red and green lines show the 5% critical boundaries. The orange vertical line marks the Bai-Perron break date of 2022-02-16. The CUSUM statistic crosses the upper boundary and remains outside the critical region for a substantial portion of the post-2022 period, confirming sustained parameter instability.

5. ML Baseline Models

Two machine learning baseline models are estimated to complement the econometric analysis and assess whether any stable predictive signal exists in the full-sample data.

Model selection and justification. OLS (Linear Regression) serves as Baseline 1, following the linear modelling tradition of Zaremba et al. (2020) and Klose (2023). Ridge Regression (L2 regularisation) serves as Baseline 2, motivated by the strong multicollinearity among regressors (DXY-US10Y: $r = 0.82$, Brent-US10Y: $r = 0.69$). A Naive Baseline (predict = 0) is included as the lower performance bound, exploiting the near-zero mean of MOEX daily returns ($\mu \approx 0.000040$). More complex nonlinear models are not estimated because the goal is diagnostic rather than predictive: if regime instability is the core problem, linear and nonlinear models will perform similarly poorly on the full sample.

Estimation and hyperparameters. The Ridge penalty α is selected via walk-forward cross-validation (`TimeSeriesSplit`, $k = 5$) on the development set (85% of data, N = 935), minimising average MAE across expanding windows. Candidates tested: $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$. Selected: $\alpha = 100.0$. The high value confirms the severity of multicollinearity. Features: Brent, DXY, VIX, US10Y, COVID_Stringency in levels. The final 15% of observations (2023-10-18 to 2024-06-14, N = 165) are reserved as a hold-out test set, untouched until final evaluation. MAPE is excluded due to its instability for near-zero return series.

Quality metrics on test set. Average metrics across 5 development folds:

Table 5: ML baseline metrics - development set (5-fold walk-forward average)

Model	**MAE**	**RMSE**	$\mathbf{R^2}$
OLS	0.01200	0.01869	−0.253
Ridge ($\alpha = 100.0$)	0.01120	0.01801	−0.156
Naive (predict = 0)	0.01068	0.01741	−0.019

Table 6: ML baseline metrics - hold-out test set (N = 165)

Model	**MAE**	**RMSE**	$\mathbf{R^2}$
OLS	0.00594	0.00844	−0.437
Ridge ($\alpha = 100.0$)	0.00562	0.00815	−0.340
Naive (predict = 0)	0.00493	0.00704	−0.000

Both linear models underperform the naive baseline on both sets (OLS MAE gap vs. naive: +0.00131 on development, +0.00101 on hold-out). The OLS train-to-hold-out gap is −0.00498, reflecting lower realised volatility in the 2024 hold-out period rather than overfitting - consistent with the post-2022 adaptation regime. Ridge improves over OLS in 4 out of 5 development folds, confirming partial multicollinearity correction.

The negative R^2 values across most folds confirm that no stable linear predictive relationship exists in the full 2020–2024 sample. This is not a model failure, but the key diagnostic finding motivating the structural break analysis: a globally fitted model cannot simultaneously represent two fundamentally different regimes. Even the Naive baseline yields R^2 near zero, consistent with the near-zero unconditional mean of MOEX returns ($\mu \approx 0.000040$) — any model introducing structurally unstable signals is penalised relative to a trivial zero-prediction benchmark. This pattern aligns with [4], who document similarly low R^2 in macro-return regressions for emerging markets during crisis periods, and reinforces the conclusion that sub-period estimation within identified regimes is the appropriate analytical strategy for this data.

Robustness analysis

Robustness is assessed through: alternative model specifications including exclusion of individual regressors - subsample estimates for each identified regime comparison of short-run and long-run coefficient directions and sensitivity of breakpoint detection to the penalty parameter (pen = 30 vs. pen = 50 yielding identical results, confirming stability).

6.7 Data Limitations

MOEX trading shutdown. The Moscow Exchange was closed from 28 February to 24 March 2022 (approximately 17 trading days).This creates a gap at the most critical moment of the shock.The Bai-Perron breakpoint at 2022-02-16 precedes the shutdown, consistent with forward-looking market behavior, but the immediate market response during the closure is unobservable.

Absence of trading volume. Trading volume for IMOEX.ME was unavailable via Yahoo Finance (all values zero). Its absence precludes any liquidity-based analysis of shock transmission, which could have provided additional evidence of regime change.

Urals-Brent divergence post-2022. From March 2022, Russian crude exports were sold at a significant discount to Brent (the Urals discount reached $30-35 per barrel at its peak). Since Brent is used throughout this study, measurement error is introduced in the commodity price regressor for the post-break period, partially explaining the weakening of the long-run Brent coefficient.

DXY as exchange rate proxy. The US Dollar Index reflects the dollar's strength against a basket of developed-market currencies rather than directly against the Russian ruble. A direct USD/RUB series was not included to avoid additional multicollinearity. This means the exchange rate channel is measured with noise.

Short sample and sub-period precision. The full analysis spans approximately 4.5 years (N = 1,102). The post-break sub-period contains approximately 566 observations - sufficient for DOLS but near the lower bound for reliable internal breakpoint detection. Cointegration test power is reduced relative to longer samples.

Non-market pricing mechanisms post-2022. After capital controls, mandatory export revenue repatriation, and exclusion of foreign investors, the MOEX index may not fully reflect fundamental market valuations. The informational content of MOEX levels in 2022-2024 is qualitatively different from the pre-2022 period, which is a structural caveat for any model treating the index as a continuous efficient-market price series throughout the full sample.

Work Plan

Table 7: Work plan for the research project

Week	Tasks	Stage of work
1-2	**Finalization of the research proposal and literature review.** Detailed study of the Bai-Perron method and `strucchange` package.	Planning
3-4	**Data collection and preparation.** Downloading daily data for MOEX Index, Brent crude oil price, USD/RUB exchange rate, VIX, and others for 01.01.2020-31.12.2025. Data cleaning, handling missing values, conducting preliminary descriptive analysis.	Data Gathering
5-6	**Cointegration analysis.** Testing time series for stationarity (ADF, KPSS tests). Assessment of the long-term relationship between the MOEX index and fundamental indicators using the dynamic OLS (DOLS) method.	Analysis
7-9	**Structural break detection.** Applying the Bai-Perron procedure to the model to determine the number and dates of structural breaks. Using supF, UDmax/WDmax tests and information criteria (BIC), also using auxiliary tests from `strucchange`.	Core Analysis
10-11	**Segmented regression and interpretation.** Estimating regression parameters for each stable regime identified by the breaks. Analyzing changes in coefficients. Interpreting structural breaks in the context of specific events (COVID-19, March 2022, sanction announcements, and others).	Analysis
12-13	**Reliability check and additional analysis.** Analyzing volatility regimes (using GARCH models). Preparing visualizations (graphs of time series with breakpoints, coefficient plots).	Validation
14-15	**Final report.** Collecting all the results, formulating conclusions, and describing limitations and future research areas.	Reporting
16	**Finalization and submission.** Final proofreading, preparation of presentation slides for the defense.	Submission

Data Limitations and Future Work

8.1 Data Limitations

MOEX trading shutdown. The Moscow Exchange was closed from 28 February to 24 March 2022 — approximately 17 trading days. This means that actual trading prices during this period are simply not available, and the market's reaction to the most acute phase of the shock cannot be directly observed. However, this does not affect the core results of the analysis — the structural break was detected at 2022-02-16, eight days before the closure, which is consistent with financial markets pricing in expected events before they formally occur. The closure therefore defines the outer boundary of the dataset, but does not affect the core of the analysis.

Sample length and sub-period precision. The original research design targeted the full period from January 2020 to December 2025, which would have allowed observation of the complete post-shock adaptation cycle including the stabilisation phase of 2023–2025. In practice, Yahoo Finance — the primary data source for this study — does not provide MOEX index data (ticker IMOEX.ME) beyond June 2024, which constrained the realised sample to 1,101 trading days. Extending coverage to 2025 would require alternative data sources such as the Moscow Exchange OpenAPI or Russian financial data providers such as Finam, which is left for future work. Within the available sample, the pre-break sub-period contains approximately 532 observations and the post-break sub-period approximately 566 observations, which is sufficient for the estimation procedures applied in this study.

Urals–Brent divergence post-2022. Throughout this study, Brent crude futures are used as the oil price variable, which is the standard choice in international finance research. From March 2022, however, Russian oil was sold at a significant discount to Brent. The Urals discount reached \$30–35 per barrel at its peak, as sanctions pushed Russia to redirect its exports to buyers who could demand lower prices in the absence of competition. This means that in the post-2022 period, the Brent price no longer perfectly reflects the oil revenues actually flowing into the Russian economy. As a result, the estimated 70% weakening of the Brent coefficient after 2022 captures the decoupling of MOEX from the global oil benchmark — which is precisely the economically relevant finding: Russian equity prices stopped tracking world oil prices in the way they used to.

Absence of trading volume. Yahoo Finance does not report trading volume for index-level tickers such as IMOEX.ME — all retrieved values were zero, making the variable unusable. Since the entire analytical framework of this study is built around price returns and macro-financial factors, this does not affect any of the models or conclusions. It does, however, mean that one potentially informative dimension — how trading activity and market liquidity changed around the 2022 shock — lies outside the scope of this study.

DXY as exchange rate proxy. The US Dollar Index reflects the dollar's strength against a basket of developed-market currencies and does not directly capture the USD/RUB exchange rate. A direct ruble exchange rate series was excluded to avoid introducing additional multicollinearity given the already high DXY–US10Y correlation ($r = 0.82$). The DXY therefore captures the broad dollar channel rather than the bilateral ruble rate, and the more than twofold strengthening of the DXY coefficient after 2022 should be interpreted as a lower-bound estimate of the exchange rate channel's increased dominance under capital controls.

Non-market pricing mechanisms post-2022. Following the introduction of capital controls, mandatory export revenue repatriation requirements, and the effective exclusion of foreign portfolio investors, the MOEX index operated under qualitatively different market microstructure conditions after 2022. This study treats MOEX as a price series and analyses its statistical relationships with macro-financial variables within each identified regime. The interpretation of results for the post-2022 period appropriately accounts for this changed institutional context.

8.2 Future Work

Extending the data horizon. The most immediate next step is to extend the MOEX dataset beyond June 2024 to cover the full originally intended period through December 2025. This would require sourcing data directly from the Moscow Exchange OpenAPI or Russian financial data providers such as Finam. A longer post-break sub-period would substantially increase the statistical power of cointegration tests within the post-2022 regime and allow observation of whether the new equilibrium relationship identified in this study is itself stable or continues to evolve as Russia's economy further adapts to sanctions.

Incorporating trading volume data. The absence of volume data points directly to a concrete extension: sourcing MOEX trading volume from the Moscow Exchange directly and using it to analyse how market liquidity changed around the 2022 shock. Given that the shock coincided with the departure of foreign institutional investors and the imposition of capital controls, volume dynamics could reveal an additional dimension of the regime shift that price-based analysis alone cannot capture — for example, whether the post-2022 market became thinner and more susceptible to price manipulation.

Replacing Brent with the Urals price series. Since Russian crude was sold at a significant discount to Brent after March 2022, replacing the Brent regressor with an actual Urals price series — available from the Russian Ministry of Energy and commodity data providers — would eliminate the measurement noise introduced by the Brent–Urals divergence. This would yield a cleaner estimate of how oil revenues actually flowing into the Russian economy relate to equity valuations, and would allow a more precise decomposition of how much of the observed 70% weakening in the Brent coefficient reflects genuine decoupling.

Improving the exchange rate specification. The DXY index captures broad dollar strength rather than the bilateral ruble rate. A natural extension is to replace or supplement DXY with a direct USD/RUB daily series, sourced from the Bank of Russia or Finam. This would allow precise identification of the ruble exchange rate channel, which the results of this study suggest became the dominant transmission mechanism after 2022. It would also enable a direct test of whether the post-2022 strengthening of the exchange rate effect reflects dollar dynamics broadly or is specific to the ruble under capital controls.

Volatility regime analysis. The 21-day rolling volatility plots clearly show two distinct volatility clusters — March 2020 and February–March 2022 — followed by a sustained low-volatility period from late 2022 onwards. Estimating a GJR-GARCH model separately for each identified regime would complement the current findings by characterising how the variance process shifted across structural breaks, not just the conditional mean. This extension would also allow a formal test of whether the asymmetric response to positive and negative shocks changed after 2022, which has direct implications for risk management on the Russian market.

Conclusion

This study set out to answer a question posed in the introduction: did the relationship between the Moscow Exchange index and its core macro-financial determinants structurally change over the period from January 2020 to mid-2024, and if so, how did the sensitivity to those determinants shift across different periods?

The answer is unambiguously yes, and it is more nuanced than a single break in February 2022. The analysis reveals that the Russian stock market passed through two distinct shocks within the sample. The first was the COVID-19 pandemic of 2020, which produced the initial volatility cluster visible in the data and established the first structural break in the long-run DOLS residuals at July 2020 — reflecting the period when markets began pricing in a prolonged disruption rather than a brief shock. The second and more fundamental shock was the geopolitical escalation of February 2022, which triggered a -40.5% single-day return, the largest in the sample, and permanently altered the transmission mechanisms linking global macro-factors to Russian equity prices.

Three hypotheses were tested, and all three are confirmed. The Chow test at 24 February 2022 ($F = 4.853$, $p < 0.0001$) and the Bai-Perron procedure together confirm that the coefficients of the short-run model changed significantly — Hypothesis 1 is supported. The DOLS cointegration analysis shows that a stable long-run equilibrium between MOEX and its macro-financial determinants exists, but only within the post-2022 regime — Hypothesis 2 is conditionally confirmed. The CUSUM test on DOLS residuals confirms that this equilibrium itself is structurally unstable across the full sample — Hypothesis 3 is confirmed.

The coefficient shifts tell a precise economic story. The sensitivity of MOEX to Brent crude oil prices fell by 70% after 2022, reflecting the growing disconnect between global oil benchmarks and the prices actually received by Russian exporters under sanctions. The exchange rate channel more than doubled in importance, consistent with capital controls making the ruble the primary transmission mechanism between the external environment and domestic equity valuations. Most strikingly, the US Treasury yield coefficient reversed sign entirely — suggesting that under financial isolation, rising global interest rates now carry a different meaning for Russian markets than they did before 2022.

The main contribution of this study is a systematic, multi-method econometric assessment of these regime shifts within a single analytical framework — something that studies focused on isolated events or single methodologies cannot provide. For investors and policymakers seeking to understand or model the current Russian market, the practical implication is clear: pre-2022 parameter estimates are no longer valid, and any model that treats the full 2020–2024 period as structurally homogeneous will produce unreliable results. These findings are bounded by the limitations and future research directions discussed in the preceding section.

The Russian stock market that existed before February 2022 — globally integrated, oil-driven, and responsive to US monetary conditions in the conventional way — is no longer the market that exists today. This study provides a quantitative account of that transformation.

References

[1] Bai, J., & Perron, P. (2003). Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1), 1-22.
URL: https://ideas.repec.org/a/jae/japmet/v18y2003i1p1-22.html (date accessed: 14.01.2026).

[2] Zeileis, A., Leisch, F., Hornik, K., & Kleiber, C. (2002). strucchange: An R Package for Testing for Structural Change in Linear Regression Models. *Journal of Statistical Software*, 7(2), 1-38.
URL: https://ideas.repec.org/a/jss/jstsof/v007i02.html (date accessed: 14.01.2026).

[3] Montalvo, J. G. (1995). Comparing cointegrating regression estimators: Some Monte Carlo evidence. *Economics Letters*, 48(3-4), 229-234.
URL: https://ideas.repec.org/a/eee/ecolet/v48y1995i3-4p229-234.html (date accessed: 14.01.2026).

[4] Zaremba, A., Kizys, R., Aharon, D. Y., & Demir, E. (2020). Infected markets: Novel coronavirus, government interventions, and stock return volatility around the globe. *Finance Research Letters*, 35, 101597.
URL: https://pubmed.ncbi.nlm.nih.gov/32550842/ (date accessed: 14.01.2026).

[5] Klose, J. (2023). Empirical effects of sanctions and support measures on stock prices and exchange rates in the Russia-Ukraine war. *Journal of International Financial Markets, Institutions and Money*, 59.
URL: https://ideas.repec.org/a/eee/glofin/v59y2024ics1044028323001205.html (date accessed: 14.01.2026).

[6] Yalcin, E., et al. (2025). On the effectiveness of the sanctions on Russia: New data and new evidence. *VoxEU/CEPR*.
URL: https://cepr.org/voxeu/columns/effectiveness-sanctions-russia-new-data-and-new-evidence (date accessed: 15.01.2026).

[7] Pan, Q., & Sun, Y. (2023). Changes in volatility leverage and spillover effects of crude oil futures markets affected by the 2022 Russia-Ukraine conflict. *Energy Economics*, 103.
URL: https://www.sciencedirect.com/science/article/abs/pii/S1544612323008140?via%3Dihub (date accessed: 14.01.2026).

[8] Atlantic Council (2025). *The Russian economy in 2025: Between stagnation and militarization*. GeoEconomics Center, Atlantic Council.
URL: https://www.atlanticcouncil.org/content-series/russia-tomorrow/the-russian-economy-in-2025-between-stagnation-and-militarization/ (date accessed: 15.01.2026).

Appendix

In the course of preparing this research project, a language model (Claude, Anthropic) was used for a limited set of technical and editorial tasks. The following prompts illustrate the nature of the assistance received.

1 LaTeX Table formatting

Model: Claude (Anthropic), claude.ai

Prompt:

"How do I create a table in LaTeX with fixed column widths using the tabular environment, so that long text wraps within cells and the table fits within the page margins with left=2.5cm, right=1.0cm geometry?"

Response summary: The model explained the use of p{width} column specifiers in the tabular environment to enable text wrapping, recommended the array and booktabs packages for cleaner formatting, and provided an example showing how to combine fixed-width columns with \setlength{\tabcolsep}{4pt} to control inter-column spacing. This guidance was applied to Tables 2 and 4 in Section 6.

2 Python visualisation: Dual-Axis time series plot

Model: Claude (Anthropic), claude.ai

Prompt:

"In Python using matplotlib, how do I plot two time series with different scales on the same figure using a secondary y-axis? I want to show MOEX index levels and Brent crude oil prices together, with vertical dashed lines marking specific dates."

Response summary: The model provided a code template using ax.twinx() to create a secondary y-axis, demonstrated how to assign each series to its respective axis with independent scaling, and showed how to add vertical reference lines using ax.axvline() with a pd.Timestamp argument. The resulting approach was used to produce Figure 4 in Section 6.4.

3 Document navigation assistance

Model: Claude (Anthropic), claude.ai

Prompt:

"I am attaching my research paper draft and the corresponding Jupyter notebook. Could you help me locate where specific numerical results appear across both files? Specifically: 1) the exact date range and observation count of the MOEX analytical dataset as confirmed by moex_data.info() output in the notebook, and 2) whether the sample period stated in the Abstract matches the date range visible in the data description section of the paper."

Response summary: The model read both the uploaded .ipynb file and the paper draft, located the relevant cell output in the notebook confirming Date range: 2020-01-06 to 2024-06-14 with Total observations: 1101, and identified a discrepancy between this output and the Abstract, which at that stage stated "January 2020 to December 2025." The model explained that the extended date range in the panel dataset reflected other indices available through Yahoo Finance, while the MOEX-specific series was limited to mid-2024 due to data provider constraints. Based on this clarification, the Abstract and Аннотация were corrected to read "January 2020 to mid-2024" throughout the final version of the paper.